

A New Era of Online Search? A Large-Scale Study of User Behavior and Personal Preferences during Practical Search Tasks with Generative AI versus Traditional Search Engines

Carolin Kaiser

Nuremberg Institute for Market Decisions
Nuremberg, Germany
carolin.kaiser@nim.org

Rene Schallner

Nuremberg Institute for Market Decisions
Nuremberg, Germany
rene@renerocks.ai

Jakob Kaiser

Nuremberg Institute for Marker Decisions
Nuremberg, Germany
jakob.kaiser@nim.org

Sabrina Schneider

Management Center Innsbruck
Innsbruck, Austria
sabrina.schneider@fhv.at

Abstract

Generative AI chatbots like ChatGPT are emerging as popular alternatives to traditional search engines for finding information online. However, their impact on user behavior and experience in everyday tasks is still unclear. We conducted a large-scale study ($n = 1,526$) with a diverse U.S. sample, using a custom-build software to track participants' behavior during practical search tasks with either ChatGPT or Google. ChatGPT users compared to Google users were faster and more likely to find the correct answers. Nevertheless, participants expressed a subjective preference for Google, with ChatGPT preference more strongly dependent on users' personality profiles. Importantly, ChatGPT users relied significantly less on primary web sources, suggesting it is more often seen as a one-stop solution rather than a connective hub for information retrieval. These findings provide key insights into the transformative effects of generative AI on everyday search, highlighting both its benefits and challenges for user interaction and trust.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; • **Applied computing** → Law, social and behavioral sciences; Psychology; • **Information systems** → Information retrieval; Users and interactive retrieval; Search interfaces.

Keywords

LLMs, Artificial Intelligence, Search Engines, Information Retrieval, User Personalities

ACM Reference Format:

Carolin Kaiser, Jakob Kaiser, Rene Schallner, and Sabrina Schneider. 2025. A New Era of Online Search? A Large-Scale Study of User Behavior and Personal Preferences during Practical Search Tasks with Generative

AI versus Traditional Search Engines. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720123>

1 Introduction

Finding relevant information quickly and reliably online is a major challenge in the modern digital age. For most of the internet era, online information searches have been dominated by keyword-based search engines like Google. However, the recent emergence of generative AI chatbots, powered by large language models (LLMs) such as ChatGPT, represents a significant shift in the landscape of information retrieval. Despite their rapid adoption and popularity, it remains an open question how generative AI tools influence the everyday user's experience when searching for information online. This study investigates this pivotal question through a large-scale experiment involving a diverse sample of U.S. residents varying in age and educational background. Specifically, we examine how the use of LLM-based chatbots compares to traditional search engines in terms of (a) users' search productivity (i.e., search accuracy and speed) b) their engagement with primary information sources, and (c) the personality traits that predict individual preferences for adopting either generative LLM tools or traditional search engines.

Generative AI refers to “computational techniques that are capable of generating seemingly new, meaningful content such as text, images, or audio from training data” [7]. Unlike traditional search engines, which provide users with a ranked list of potentially relevant websites based on keyword queries, LLM chatbots dynamically generate text responses to user prompts. Since the introduction of ChatGPT by OpenAI in November 2022, the adoption of generative AI technologies has accelerated rapidly, with ChatGPT reaching its first million users within just five days [3, 22]. Notably, while developers of LLM chatbots increasingly offer features allowing for direct web search via their platform, traditional search engines are integrating generative AI features into their standard search results [8, 16]. Thus, there is currently a trend both for technology companies and their customers to complement, or in some cases maybe even replace traditional search engines with generative AI chatbots for finding information online [27].

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3720123>

This ongoing shift in user preferences provides a unique opportunity for the field of human-computer interaction (HCI) to study the preconditions and implications of large-scale technology adoption in the general population. From a functional perspective, the acceptance of new technologies often depends on associated efficiency gains [24]. Indeed, prior research has demonstrated that LLMs enhance productivity and efficiency in text-generation tasks, such as drafting or revising reports [6, 15]. However, it remains unclear whether such benefits extend to the domain of online information retrieval [9]. Searching for information online does not primarily demand the generation of text, but rather the accurate identification of the most relevant information related to the current search problem. Therefore, our study measured both the effectiveness and efficiency of participants performing a practical search task using either ChatGPT or Google, as currently prevalent examples for these different approaches to retrieve information online.

The implications of LLM systems for online search might extend beyond mere task performance. Current AI chatbots and search engines differ fundamentally in how they facilitate engagement with online sources. While search engines traditionally serve as hubs between users and external websites, LLM chatbots generate self-contained responses by synthesizing information from web content. This distinction raises important questions: Do LLMs encourage or discourage direct interaction with primary sources? Could their use lead to a decline in fact-checking or an increased risk of misinformation due to the known tendency of LLMs to “hallucinate” inaccurate responses [11]? To explore these possibilities, we quantified how reliance on LLMs versus search engines influenced users’ willingness to engage with multiple information sources online.

Finally, while objective performance differences are important, the adoption of new technologies also depends on individual personality traits [4, 12]. Using the Big Five Inventory, a well-established framework for assessing five core personality traits (extraversion, openness, conscientiousness, agreeableness, and neuroticism [17, 18, 26]), we examined whether personality predicts preferences for using LLM-based search or classic search engines. While exploratory in nature, this analysis might offer valuable insights into the personality-specific determinants of technology adoption, shedding light on why some individuals may embrace LLM-based tools while others do not [2].

Overall, the goal of this study was an empirical, large-scale comparison of the differences in user behavior and user experience when solving a practical online search task with either ChatGPT (as the currently prevalent LLM Chatbot) or Google (the currently prevalent search engine). For this purpose, we designed an experiment where users had to engage with a controlled, but reasonably realistic search scenario that they might encounter in everyday life, while their interactions with either search tool were tracked using purpose-built software. More specifically, while online search can include a host of different scenarios and affordances [20], we opted for a task where participants had to identify the optimal choice in different domains (either finding a new smartphone or a holiday destination) based on predefined criteria, allowing for a quantifiable comparison of search effectiveness and efficiency. Post-task measures captured participants’ willingness to use either ChatGPT or Google for future tasks, as well as their Big-5 personality profiles. By integrating these data, we contribute to the understanding of

human-computer interactions by exploring (a) the effectiveness and efficiency of LLM-based search compared to traditional search engines, (b) their impact on user engagement with diverse information sources (i.e., other websites), and (c) the role of personality in shaping technology adoption. We thus provide an experimentally controlled, large-scale insight into the transformation of search behavior due to the rise of generative AI in everyday life.

2 Methods

2.1 Participants

Overall, 1,924 people from the USA took part in an online study. Participant acquisition was performed via Prolific and was representative with respect to gender and age (excluding minors) to current US census data. Participants received the dollar equivalent of 4.50 pounds (approx. \$ 5.50) as a guaranteed reimbursement. Additionally, to ensure task motivation, participants were paid a performance-dependent reward equivalent of 0.50 pounds if they performed the search task successfully (s. below). Due to lack of task adherence (see below for precise criteria), we excluded 398 participants from the data analysis, resulting in a final sample of 1,526 participants. Demographic information of our study sample is shown in Table 1. The experimental groups did not differ from each other with regards to age, educational attainment or gender distribution (all p ’s $> .10$).

2.2 Procedure

Participants performed a practical research task that necessitated to find information online to select the optimal option among several alternatives, either by using a classic search engine (Google) or an LLM chatbot (ChatGPT). This task was performed on a custom-built browser application, which allowed us to track participants’ search behavior and speed.

2.2.1 Technical Setup. A custom browser software for desktop computers was developed to work alongside a secure server backend, enabling the tracking of behavior in a realistic online environment. The browser application was developed entirely from scratch, embedding a Chromium-based web engine to ensure maximum compatibility with existing websites. This approach enabled participants to interact seamlessly with platforms such as ChatGPT, Google, and any additional websites visited during their research, thereby preserving an authentic user experience without functional limitations. This application offered the essential functionality of a typical browser, including an address bar, website display, and the capability to open multiple tabs. The research tasks and subsequent questionnaires appeared on the left-hand side, while the browser window was positioned on the right-hand side of the screen. Copying and pasting of the instructions was disabled to encourage participants to engage actively with the search task. During the task, the application recorded users’ text inputs, URLs of visited websites, and engagement duration with each site. Meanwhile, a server backend managed the experiment workflow by handling the task presentation and securely storing collected user input upon task completion. Participants were informed beforehand that their data would be recorded. To protect participants’ privacy, the application never stored or transmitted any passwords or email addresses.

Table 1: Sociodemographic Descriptives for each Experimental Group

	ChatGPT Smartphone	ChatGPT Theme Park	Google Smartphone	Google Theme Park	Overall Sample
Sample Size	412	382	359	373	1526
Gender					
Male	50.2%	49.5%	50.4%	45.0%	48.8%
Female	49.8%	49.7%	49.0%	54.7%	50.8%
Other	0.0%	0.8%	0.6%	0.3%	0.4%
Age					
18-29	26.2%	30.9%	31.2%	30.0%	29.5%
30-44	45.6%	41.4%	41.2%	40.5%	42.3%
45-59	20.4%	20.9%	21.7%	22.0%	21.2%
60+	7.8%	6.8%	5.8%	7.5%	7.0%
Educational Level					
High School or Less	12.1%	12.0%	13.4%	13.4%	12.7%
College/Associate Deg.	36.2%	29.6%	38.4%	36.2%	35.1%
Bachelor's Degree	33.7%	37.7%	34.5%	35.4%	35.3%
Advanced Degree	18.0%	20.7%	13.6%	15.0%	16.9%

After the experiment, data recording was halted, and participants were instructed on how to uninstall the application. During the task, participants used the publicly available versions of the Google platform and ChatGPT in the USA at the time of the data collection in October/November 2024.

2.2.2 Experimental Procedure. Once participants had started the application, they were asked to imagine helping a friend finding the best product among three alternatives according to a list of three criteria. More specifically, each participant performed one of two tasks: 1) In the smartphone task, participants had to identify out of three phones (ASUS Zenfone 8, Motorola Moto E7 Plus, Samsung Galaxy A72) the one which had a) at least 4,250 mAh battery capacity, b) a camera with minimum resolution of 64 MP, c) at least 6 GB RAM. 2) In the theme park task, participants had to select one of three theme parks for a family trip (Six Flags Magic Mountains, Disneyland Resort, Legoland) which had a) at least 60 attractions, b) an overall size of at least 220 acres, and c) availability of seasonal events for Christmas. We included two different topics in our experiment to enhance the generalizability of our results to different knowledge domains, while employing the same task structure (i.e., identifying the optimal choice based on predefined criteria) in each case. Pilot testing of the task had indicated that while for technological products like smartphones information of features are more widely available online (making this type of search potentially easier), identifying features of individual theme parks required more detailed research. For both search topics, only one option fulfilled all criteria (smartphone: Samsung Galaxy A72, theme park: Disneyland Resort). Participants were informed that they would receive a bonus payment of the dollar equivalent of 50 pence after study completion if they identified the correct option. Importantly, we randomly assigned each participant to one of two search tools. They either were instructed to employ Google (as the currently prevalent search engine) or ChatGPT (as the currently prevalent LLM Chatbot) for their search. Participants were allowed,

but not required to, visit other websites during their search in addition to the assigned search tool. Participants were free to search for the best options for as long as they wished. Once they were satisfied that they had gained sufficient information, they pressed a designated button on the screen. Subsequently, participants indicated their choice from a list of all three available phones/holiday destinations. They received no feedback about the correctness of their choice. Subsequently, participants filled out a battery of questionnaires, which included general demographic information, their intention to use the assigned search tool in the future (measured on a 7-point scale previously developed and validated, [24]) and a previously validated version of the Big-5 personality inventory [17].

2.3 Data Quality Assurance and Statistical Evaluation

We conducted several data quality checks to ensure that participants read and followed the task instructions. First, the post-task questionnaire included two attention check items where participants were explicitly instructed to select a specific response. Participants who failed to do so were deemed unlikely to have read the questions and were excluded ($n = 61$, 3.17% of total sample). Participants were also removed if browsing data indicated that they never used their assigned search tool ($n = 112$, 5.82%), or if they input no or minimal text in the browser (fewer than 20 characters, $n = 133$, 6.91%). Lastly, we removed participants with suspiciously low search durations (less than 1 minute; $n = 64$, 3.33 %) or exceptionally large search durations (more than 3 standard deviations away from sample mean; $n = 28$, 1.46 %). After applying these quality control measures, the final sample comprised 1,526 participants, who likely engaged meaningfully with the assigned task. To ensure the validity of subsequent statistical tests, we assessed relevant assumptions, including the normality of distributions and homogeneity of variances, for each test. We report the p-values of robust test alternatives in case where violations of these assumptions would

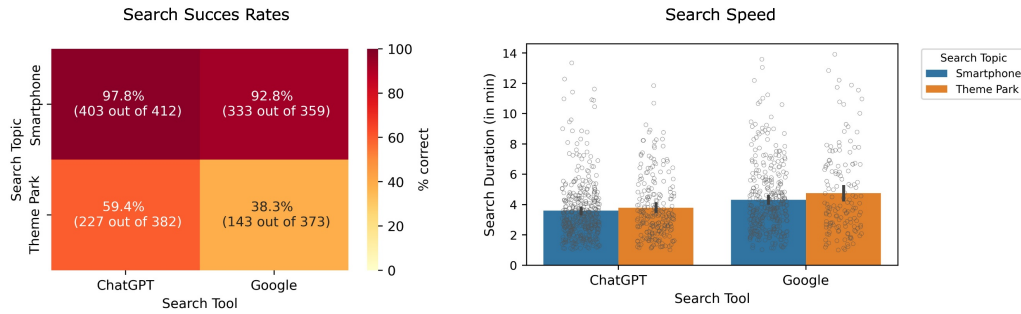


Figure 1: Search performance depending on search tool and search task. Left: Success rates in each group (with total number of participants per condition in parentheses). Right: Average durations for successful searches. Dots represent individual participants.

have led to different statistical outcomes with standard tests. Correlational analyses were conducted with bootstrapped confidence intervals (1,000 iterations) and controlled for family-wise error rates to support statistically reliable conclusions.

3 Results

3.1 Tool-Dependent Differences in Search Performance

Figure 1 shows the search performance of participants in terms of success rates and search durations separately for the two search tools (ChatGPT/Google) and the two search topics (smartphones/theme parks). Regarding success rates, chi-square analysis indicated that for both tools, smartphone searches compared to theme park searches were more likely to be successful, for ChatGPT $\chi^2(1) = 175.93, p < .001$; for Google: $\chi^2(1) = 235.84, p < .001$. Thus, as expected, finding the best-fitting product online was harder for a non-technological topic than a technological topic. Importantly, for both search topics, the ChatGPT group compared to the Google group had a significantly higher success rate, for smartphone: $\chi^2(1) = 10.19, p = .001$; for theme park: $\chi^2(1) = 32.74, p < .001$. Thus, ChatGPT users compared to Google users were more likely to find the correct answer in both search tasks.

We compared the search times of all successful participants to evaluate the efficiency of the search tools for the experimental task (cf. Figure 1). A between-subject ANOVA with the factors TOOL (ChatGPT/Google) and TOPIC (Smartphone/Theme Park) revealed a main effect of TOPIC, $F(1, 1102) = 4.45, p = .04, \eta^2_p = .004$, indicating that smartphone searches were faster than searches for the optimal theme park. Furthermore we found a main effect of TOOL, $F(1, 1102) = 36.05, p < .001, \eta^2_p = .032$, showing that ChatGPT users found the correct solution faster than Google users. There was no TOOL x TOPIC interaction, $F(1, 1102) = 0.81, p = .37, \eta^2_p = .001$. ChatGPT users compared to Google users found the correct answer faster both for the smartphone search ($p < .001, d = 0.30$), and the theme park search ($p = .006, d = 0.37$). Thus, overall, LLM use compared to a traditional search engine led to higher effectiveness and efficiency.

3.2 Tool-Dependent Change in Search Behavior

Participants were free to use any other websites in addition to their primary search tool (ChatGPT or Google). To test if the use of LLMs compared to search engines influenced browsing behavior, we evaluated the visits of external websites, here defined as any websites that users interacted with during their search except Google and ChatGPT (Figure 2). For this analysis, we differentiated between successful and unsuccessful searches to evaluate the effectiveness of website use. (The topic of search had no significant influence on the observed pattern of results and, thus, is not further discussed.) A between-subject ANOVA with the factors TOOL (ChatGPT/Google) and OUTCOME (Success/Failure) revealed a main effect of TOOL, $F(1, 1522) = 632.23, p < .001, \eta^2_p = .29$, and a main effect of OUTCOME, $F(1, 1522) = 23.03, p < .001, \eta^2_p = .02$. The use of ChatGPT compared to Google as primary search tool led to a significant decline in the visit of other websites to find the correct answer, both for successful searchers ($p < .001, d = 1.60$) and unsuccessful searchers ($p < .001, d = 0.76$). In fact, the vast majority of ChatGPT users (87.78 %) did visit no external websites at all (for Google users: 26.91 %). We also found a significant TOOL x OUTCOME interaction, $F(1, 1522) = 19.76, p = .004, \eta^2_p = .01$, which indicated that the impact of users' website visits on their task success was dependent on their primary search tool. While successful compared to unsuccessful Google users visited significantly more websites during search, ($p < .001, d = .37$), for ChatGPT users the number of websites visited had no impact on their task success ($p = .38, d = .03$). Thus, ChatGPT users compared to Google users visited far fewer external websites, without this negatively impacting their search performance.

3.3 Subjective Search Tool Preference

A key question to understand human-computer interaction is in how far the adaptation of new technologies depends on individual's personalities. We asked participants to rate their intention to use the assigned search tool in the future, and, for participants successful in the task (to avoid a bias to due different success rates depending on the assigned search tool, s. above), we calculated the correlations between individuals' tool ratings and their personality facets as measured with the Big-5 inventory. Independent of their search topic, participants indicated higher likelihood to use Google

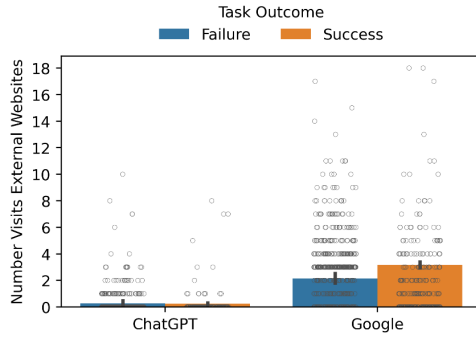


Figure 2: Number of external websites (i.e., websites other than the assigned search tool) visited by users during the search task. Dots represent data of individual participants.

than ChatGPT for future searches (smartphone group: $p < .001$, $d = 0.86$, theme park group: $p < .001$, $d = 0.79$). Thus, despite ChatGPT leading to a better and faster performance in the current task, users indicated a subjective preference for Google. Analysis of distribution variances (cf. Figure 3) indicated that usage intention varied significantly strongly for ChatGPT compared to Google, for which almost all participants reported a high usage intention (Leven's test: $W = 170.66$, $p < .001$; cf. Figure 3). This would indicate that the use of LLM tools compared to Google varies more depending on individual user characteristics. This was supported by our correlational analysis: For Google, we found no significant relationship between users' Big 5 personality traits and their subjective evaluations (all p 's $> .10$, cf. Figure 3 for individual values and confidence intervals). In contrast, ChatGPT usage intention correlated with several factors: Higher extroversion, conscientiousness, agreeableness and openness to experience predicted higher intention to reuse ChatGPT (all p 's $< .02$). Higher neuroticism (which describes individuals more prone to skepticism and concern) predicted less intention to reuse ChatGPT ($r = -0.22$, $CI = [-.30, -.14]$, $p < .001$). This indicates that traditional web search tools like Google are widely accepted, meaning that most users across the board intend to use them. In contrast,

willingness to use LLM-based search shows stronger variance and depends more strongly on users' personality profiles.

4 Discussion

This study investigated the impact of generative AI chatbots (here: ChatGPT) compared to traditional search engines (here: Google) on search behavior and subjective evaluations during typical everyday search scenarios. Using a large-scale experimental design with a diverse sample of U.S. participants, our findings revealed significant differences in search performance, engagement with external sources, and personality-specific tool preferences. These results contribute to our understanding of the rapid adoption of generative AI technologies and highlight potential challenges for their integration into daily life from a human-computer interaction perspective.

Our results demonstrated that ChatGPT enabled faster and more accurate searches compared to Google. This finding aligns with prior evidence suggesting that LLM-based tools enhance productivity in text-focused tasks, such as writing and editing [6, 15]. By extending these observations to online search, this study supports the claim that LLM tools can boost productivity in practical tasks. It should be emphasized that our results do not indicate that ChatGPT will always lead to better search outcomes. For example, to maintain feasibility, we provided users with clearly defined search criteria that could be reasonably well evaluated within one online session. Future research should examine whether these performance benefits persist in more complex or open-ended search scenarios, where users must synthesize more extensive information.

A notable behavioral distinction emerged in the way participants engaged with external websites. Users of ChatGPT relied exceedingly less on supplementary web sources compared to Google users, suggesting that ChatGPT's summaries often sufficed to meet their search goals. Given that traditional search engine like Google primarily present a list of websites (and smaller text excerpts from those sources), it stands to reason that typical Google-based searches involve at least some website visits. Importantly, however, there is to date little data about how LLM-based tools influence typical patterns of website visits, for example to verify or supplement AI-generated summaries of information. Our study clearly

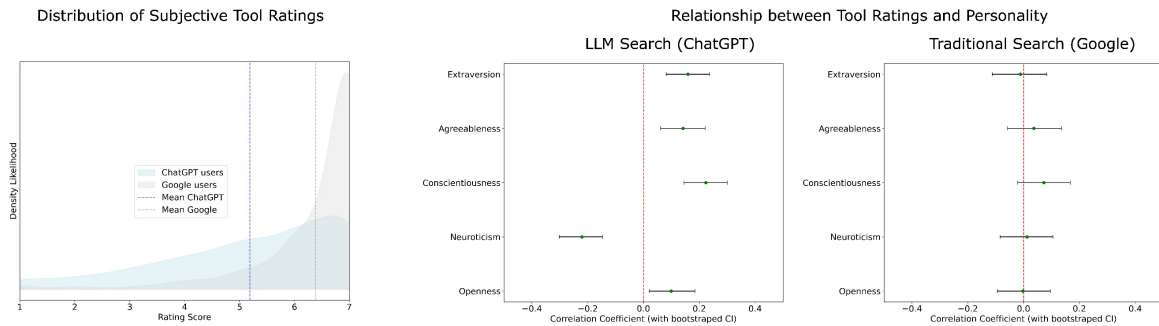


Figure 3: Left: Probability distribution of self-reported intention to use ChatGPT/Google in the future, indicating a markedly stronger variance in usage intention for ChatGPT compared to Google. Right: Tool-dependent correlations between usage intention and personality factors. Dots represent estimated correlation coefficient. Bars represent bootstrapped confidence intervals (CI) with 1000 iterations each. Confidence intervals crossing 0 represent nonsignificant correlations.

shows that tools like ChatGPT not only can lead to significantly less but, in a majority of cases in the current study, no visits of other websites at all. While this convenience helps to explain the growing popularity of generative AI tools, it also raises important concerns. Reduced engagement with external websites could negatively impact platforms reliant on user traffic, such as product review sites or journalistic outlets. Moreover, this behavior could increase users' susceptibility to potential inaccuracies in AI-generated summaries, such as misinformation resulting from LLM hallucinations or biases in training data [5, 11]. Although these risks are relatively low in consumer-oriented scenarios such as investigated in this study (e.g., shopping for a phone), the consequences could be more severe in domains requiring high-stakes decision-making, such as healthcare or finance [13]. Future studies should explore whether users' reliance on generative AI summaries diminishes when tasks involve higher stakes or greater complexity, which might motivate more active engagement with diverse sources.

Interestingly, despite ChatGPT's performance benefits, we found that Google users reported a higher subjective willingness to reuse this tool for future searches. Analysis of personality differences revealed that this stemmed from a higher variance in ChatGPT usage intention: While Google was perceived as a highly relevant tool by the vast majority of participants, intentions to use ChatGPT were more varied, i.e., dependent on participants' personalities. While the discussion of all personality traits is beyond the scope of this report, it seems noteworthy that our data indicates that the main personality trait leading to lower LLM adoption (but not less Google use) was neuroticism. The negative relationship between neuroticism and ChatGPT might stem from the relative newness of this tool and related uncertainty. Since neuroticism is associated with a general pessimistic outlook [18, 19], it stands to reason that people higher in neuroticism are less likely to be among early adopters for new technologies [2]. Here, designers of LLM tools could support wider user acceptance by improving safety features and reliability of their software to lower reluctance in more cautious users.

The current study integrates with and extends previous theoretical and empirical work regarding changes in search behaviour due to LLM chatbots, for example in the areas of healthcare [1], scientific research [10], image search [25] and consumer products [21]. Importantly, while most studies found that LLM chatbots can increase search efficiency, their effect on search effectiveness or outcome quality have been mixed. This suggests that the beneficial and detrimental effects of LLM chatbots on information retrieval might depend on the task goal and context. It is important to note that, while we employed two different search topics to enhance generalizability of our findings, the structure of the two tasks employed here (selecting an optimal option based on predefined criteria) were identical and by no means present a comprehensive account of all types of online search. Previous studies suggested systematizations of different types of search scenarios [14, 20, 23]. Based on such systematic approaches, for future studies, it would be important to compare LLM-based and traditional search tools with regards to search task that, for example, accounts for different types of search affordances and strategies.

A further limitation of our study was that we asked participants to solve imaginary search tasks. While we carefully selected scenarios that arguably resemble typical online searches in real life, it is likely that participants engaged with a lower amount of motivation compared to their real-life decision making. We chose this approach because it allowed for a systematic comparison of participants solving the same task with different tools in a controlled experiment. Furthermore, as our performance measure indicate, there was sufficient motivation to successfully solve the provided tasks. Nevertheless, future research could employ field studies to observe participants during their naturally occurring browsing behavior.

To conclude, our study delivers clear empirical evidence that AI tools compared to traditional search engines can improve effectiveness and efficiency in typical everyday search scenarios. Nevertheless, our data also shows that generative AI tools are not as widely used and accepted as typical search engines, with AI tool use being more dependent on the specific personality profile of the user. Moreover, our data strongly suggests that the increased reliance on AI-generated summaries can massively reduce engagement with external websites, posing potential challenges for online platforms dependent on user traffic and potentially raising the risk of misinformation. By examining the interplay between tool performance, engagement behavior, and individual personality traits, this research offers valuable insights for designers and users of generative AI tools. As these technologies continue to shape everyday information retrieval, understanding their benefits and risks will be critical to guiding their responsible integration into society.

References

- [1] Amir Reza Ashraf, Tim Ken Mackey, and András Fittler. 2024. Search Engines and Generative Artificial Intelligence Integration: Public Health Risks and Recommendations to Safeguard Consumers Online. *JMIR Public Health and Surveillance* 10: e53086. <https://doi.org/10.2196/53086>
- [2] Tim Barnett, Allison W Pearson, Rodney Pearson, and Franz W Kellermanns. 2015. Five-factor model personality traits as predictors of perceived and actual usage of technology. *European Journal of Information Systems* 24, 4: 374–390. <https://doi.org/10.1057/ejis.2014.10>
- [3] Alexander Bick, Adam Blandin, and David J Deming. 2024. *The rapid adoption of generative ai*. National Bureau of Economic Research.
- [4] Cinzia Calluso and Maria Giovanna Devetag. 2024. The impact of technology acceptance and personality traits on the willingness to use AI-assisted hiring practices. *International Journal of Organizational Analysis*. <https://doi.org/10.1108/IJOA-06-2024-4562>
- [5] Sophia Cheng. 2024. When Journalism Meets AI: Risk or Opportunity? *Digital Government: Research and Practice*: 3665897. <https://doi.org/10.1145/3665897>
- [6] Fabrizio Dell'Acqua, Edward McFowland, Ethan R. Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Saran Rajendran, Lisa Kraye, François Candelon, and Karim R. Lakhani. 2023. Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4573321>
- [7] Stefan Feuerriegel, Jochen Hartmann, Christian Janiesch, and Patrick Zschech. 2024. Generative AI. *Business & Information Systems Engineering* 66, 1: 111–126. <https://doi.org/10.1007/s12599-023-00834-7>
- [8] Google. 2024. Generative AI in Search: Let Google do the searching for you. Retrieved January 21, 2025 from <https://blog.google/products/search/generative-ai-google-search-may-2024/>
- [9] Vinayaka Gude. 2025. Factors Influencing ChatGpt Adoption for Product Research and Information Retrieval. *Journal of Computer Information Systems*: 1–10. <https://doi.org/10.1080/08874417.2023.2280918>
- [10] Yong Nam Gwon, Jae Heon Kim, Hyun Soo Chung, Eun Jee Jung, Joey Chun, Serin Lee, and Sung Ryul Shim. 2024. The Use of Generative AI for Scientific Literature Searches for Systematic Reviews: ChatGPT and Microsoft Bing AI Performance Evaluation. *JMIR Medical Informatics* 12: e51187–e51187. <https://doi.org/10.2196/51187>

- [11] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2024. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Transactions on Information Systems*: 3703155. <https://doi.org/10.1145/3703155>
- [12] Sawsen Lakhal and Hager Khechine. 2017. Relating personality (Big Five) to the core constructs of the Unified Theory of Acceptance and Use of Technology. *Journal of Computers in Education* 4, 3: 251–282. <https://doi.org/10.1007/s40692-017-0086-5>
- [13] Sarah Lebovitz, Hila Lifshitz-Assaf, and Natalia Levina. 2022. To Engage or Not to Engage with AI for Critical Judgments: How Professionals Deal with Opacity When Using AI for Medical Diagnosis. *Organization Science* 33, 1: 126–148. <https://doi.org/10.1287/orsc.2021.1549>
- [14] Jiqun Liu. 2021. Deconstructing search tasks in interactive information retrieval: A systematic review of task dimensions and predictors. *Information Processing & Management* 58, 3: 102522. <https://doi.org/10.1016/j.ipm.2021.102522>
- [15] Shakked Noy and Whitney Zhang. 2023. Experimental evidence on the productivity effects of generative artificial intelligence. *Science* 381, 6654: 187–192. <https://doi.org/10.1126/science.adh2586>
- [16] OpenAI. 2024. Introducing ChatGPT with Search. Retrieved January 21, 2025 from <https://openai.com/index/introducing-chatgpt-search/>
- [17] Beatrice Rammstedt. 2007. The 10-Item Big Five Inventory. *European Journal of Psychological Assessment* 23, 3: 193–201. <https://doi.org/10.1027/1015-5759.23.3.193>
- [18] Brent W. Roberts, Nathan R. Kuncel, Rebecca Shiner, Avshalom Caspi, and Lewis R. Goldberg. 2007. The Power of Personality: The Comparative Validity of Personality Traits, Socioeconomic Status, and Cognitive Ability for Predicting Important Life Outcomes. *Perspectives on Psychological Science* 2, 4: 313–345. <https://doi.org/10.1111/j.1745-6916.2007.00047.x>
- [19] Michelle Nadine Servaas, Harriëtte Riese, Johan Ormel, and André Aleman. 2014. The neural correlates of worry in association with individual differences in neuroticism. *Human Brain Mapping* 35, 9: 4303–4315. <https://doi.org/10.1002/hbm.22476>
- [20] Chirag Shah and Emily M. Bender. 2022. Situating Search. In *ACM SIGIR Conference on Human Information Interaction and Retrieval*, 221–232. <https://doi.org/10.1145/3498366.3505816>
- [21] Sofia Eleni Spatharioti, David Rothschild, Daniel G Goldstein, and Jake M Hofman. 2025. Effects of LLM-based Search on Decision Making: Speed, Accuracy, and Overreliance.
- [22] Statista. 2023. Threads shoots past one million user mark at lightning speed. Retrieved January 21, 2025 from <https://cdn.statcdn.com/Infographic/images/normal/29174.jpeg>
- [23] Pertti Vakkari. 2016. Searching as learning: A systematization based on literature. *Journal of Information Science* 42, 1: 7–18. <https://doi.org/10.1177/0165551515615833>
- [24] Venkatesh, Thong, and Xu. 2012. Consumer Acceptance and Use of Information Technology: Extending the Unified Theory of Acceptance and Use of Technology. *MIS Quarterly* 36, 1: 157. <https://doi.org/10.2307/41410412>
- [25] Albatool Wazzan, Stephen MacNeil, and Richard Souvenir. 2024. Comparing Traditional and LLM-based Search for Image Geolocation. In *Proceedings of the 2024 ACM SIGIR Conference on Human Information Interaction and Retrieval*, 291–302. <https://doi.org/10.1145/3627508.3638305>
- [26] Lisa Weik, Leonard Fehring, Achim Mortsiefer, and Sven Meister. 2024. Big 5 Personality Traits and Individual- and Practice-Related Characteristics as Influencing Factors of Digital Maturity in General Practices: Quantitative Web-Based Survey Study. *Journal of Medical Internet Research* 26: e52085. <https://doi.org/10.2196/52085>
- [27] ChengXiang Zhai. 2024. Large Language Models and Future of Information Retrieval: Opportunities and Challenges. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 481–490. <https://doi.org/10.1145/3626772.3657848>